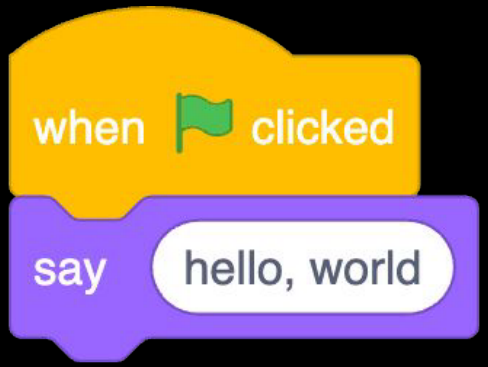
This is CS50

GETTING INFORMATION
FROM INET
IS LIKE TRYING TO
GET A DRINK
FROM A
FIREHOSE


when clicked
say
hello, world

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

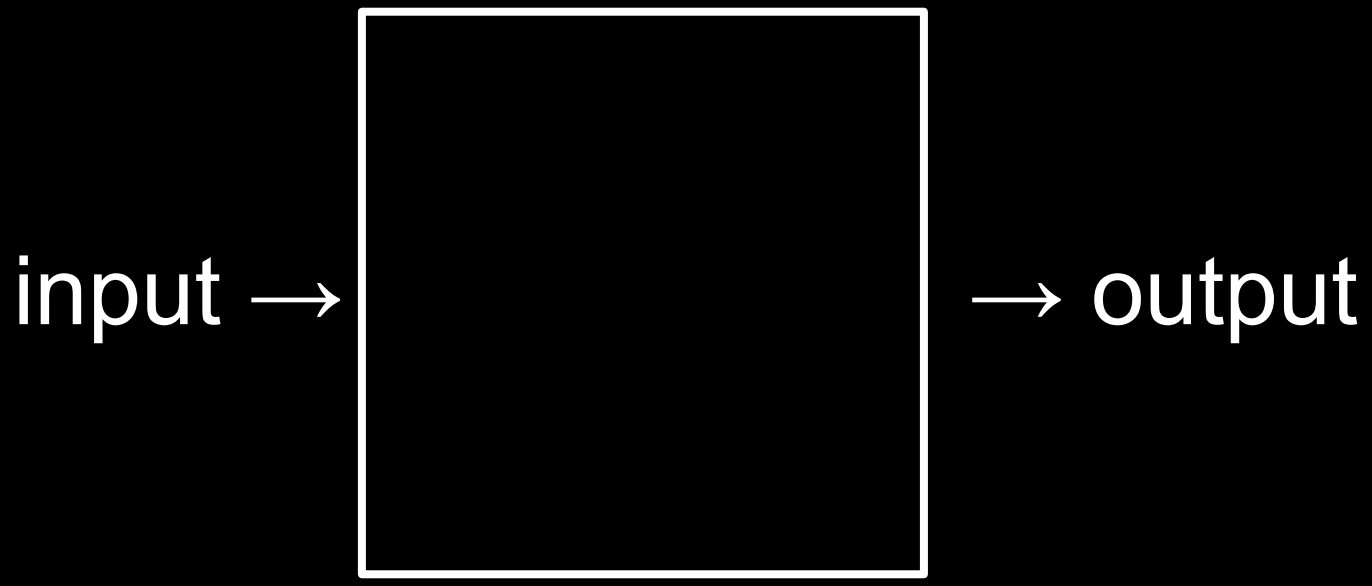
source code

```
01111111 01000101 01001100 01000110 00000010 00000001 00000001 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000010 00000000 00111110 00000000 00000001 00000000 00000000 00000000
10110000 00000101 01000000 00000000 00000000 00000000 00000000 00000000
01000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
11010000 00010011 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 01000000 00000000 00111000 00000000
00001001 00000000 01000000 00000000 00100100 00000000 00100001 00000000
00000110 00000000 00000000 00000000 00000101 00000000 00000000 00000000
01000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
01000000 00000000 01000000 00000000 00000000 00000000 00000000 00000000
01000000 00000000 01000000 00000000 00000000 00000000 00000000 00000000
11111000 00000001 00000000 00000000 00000000 00000000 00000000 00000000
11111000 00000001 00000000 00000000 00000000 00000000 00000000 00000000
00001000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000011 00000000 00000000 00000000 00000100 00000000 00000000 00000000
00111000 00000010 00000000 00000000 00000000 00000000 00000000 00000000
...
```

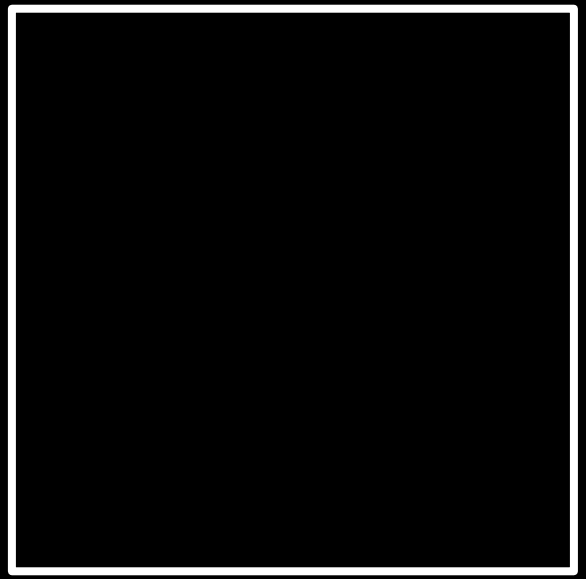
machine code

0

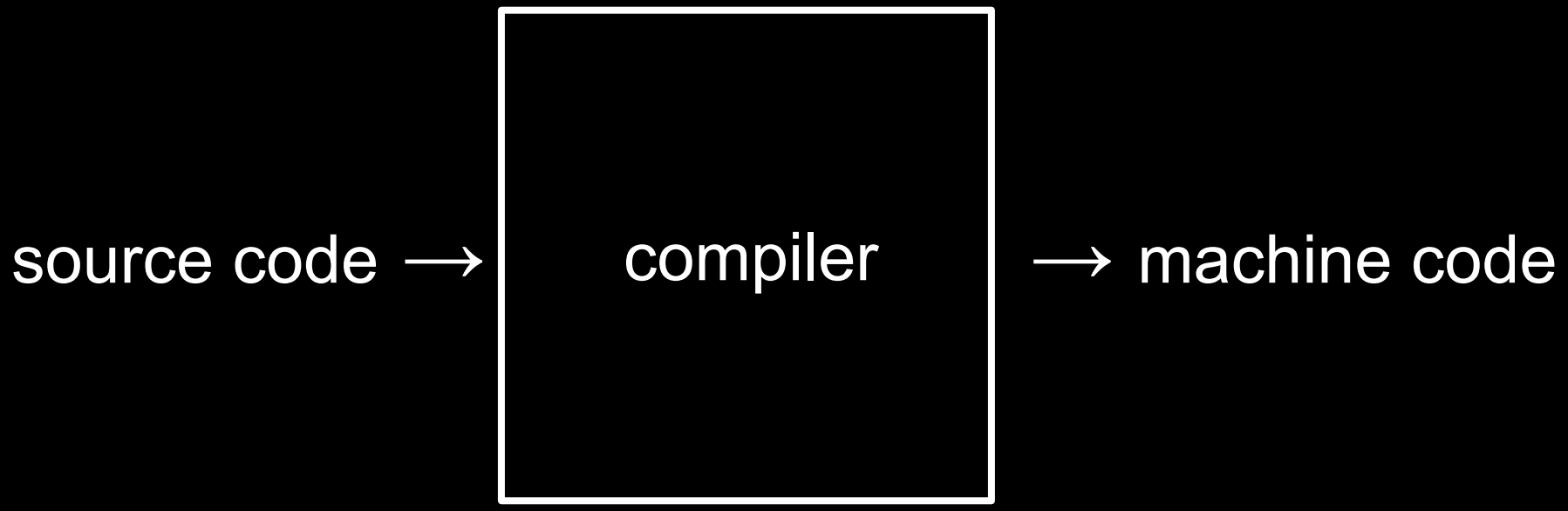
1


input →
→ output

source code $\longrightarrow$



source code $\longrightarrow$ $\longrightarrow$ machine code


source code
compiler
machine code

Visual Studio Code for CS50

cs50.dev

EXPLORER

HELLO [CODESPACES]

hello.c

hello.c

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

TERMINAL

```
$ make hello
```

OUTLINE

TIMELINE

EXPLORER

HELLO [CODESPACES]

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hello.c

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EXPLORER

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EXPLORER

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hello.c

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```c
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int main(void)
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}
```

TERMINAL

```
$ make hello
```

OUTLINE

TIMELINE

graphical user interface

GUI

EXPLORER

HELLO [CODESPACES]

hello.c

hello.c

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

TERMINAL

```
$ make hello
```

OUTLINE

TIMELINE

command-line interface

CLI

EXPLORER

HELLO [CODESPACES]

hello.c

hello.c

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

TERMINAL

```
$ make hello
```

OUTLINE

TIMELINE

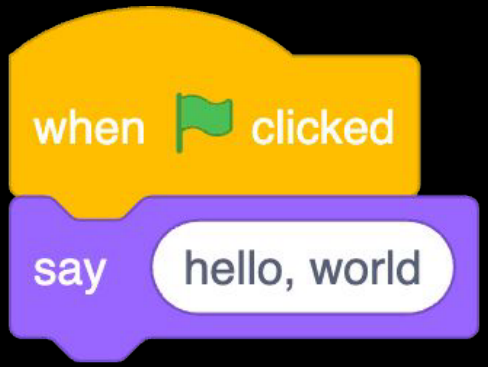
```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

```
code hello.c

make hello

./hello
```


when clicked
say
hello, world

```c
#include <stdio.h>

int main(void)
{
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}
```
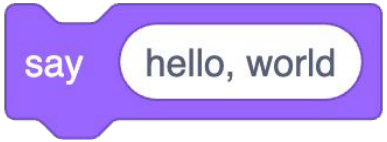
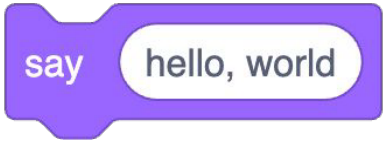
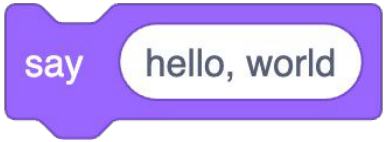


say
hello, world

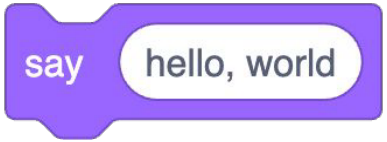



```
print (                    )
```

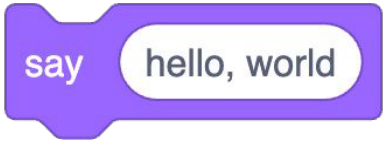



```
printf(                    )
```

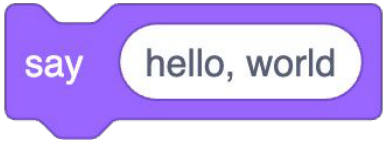
format



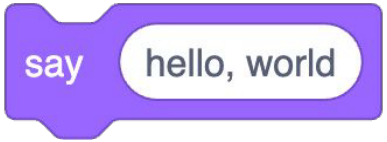

```
printf( hello, world   )
```




```
printf("hello, world  ")
```

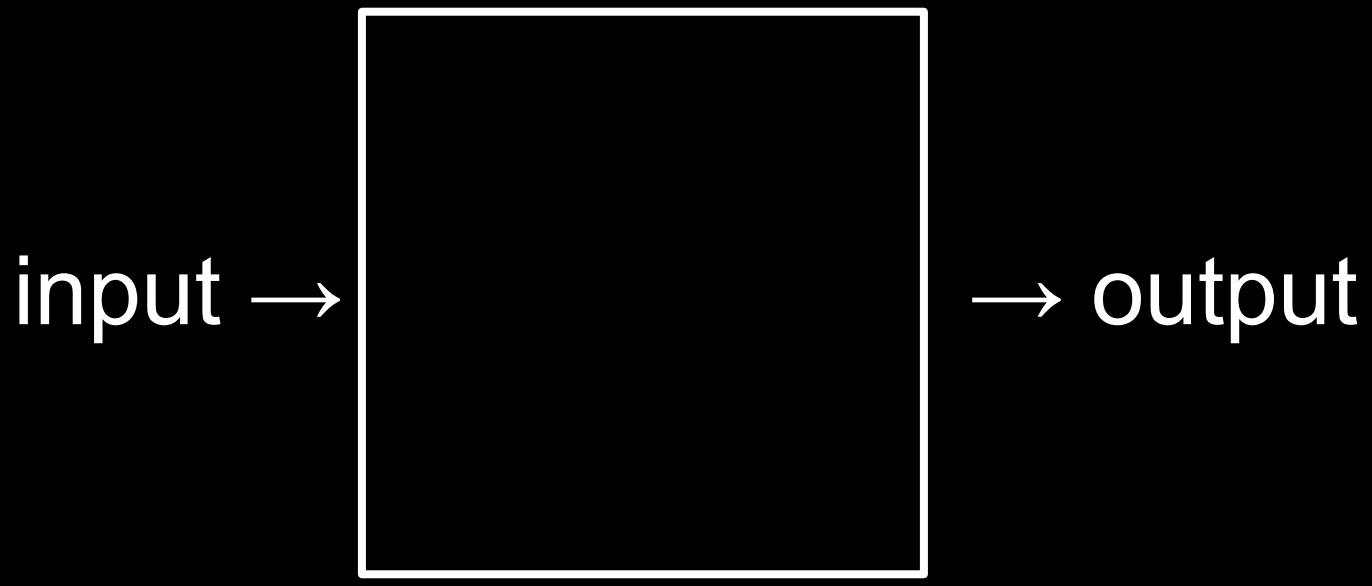



```
printf("hello, world\n")
```

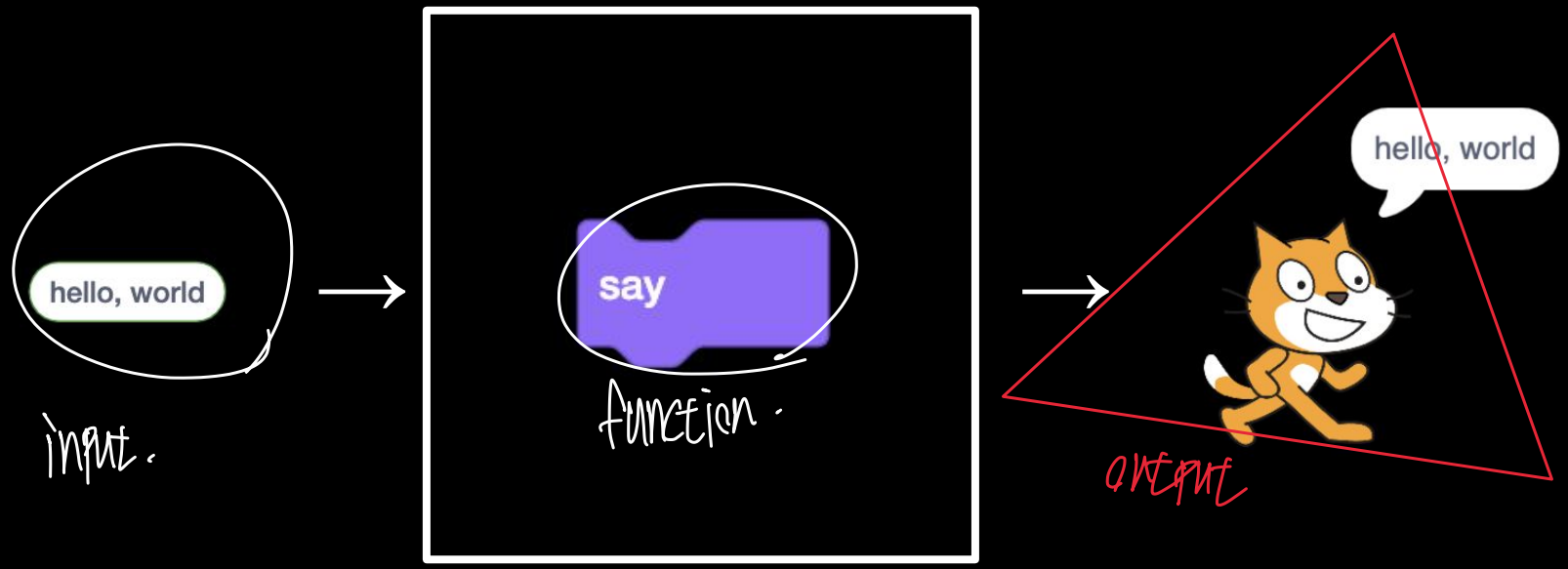



```
printf("hello, world\n");
```

$$f(x)$$


input →
→ output

arguments $\longrightarrow$ function $\longrightarrow$ side effects


hello, world
input.
say
function.
hello, world
output

escape sequences

MUST backslash.

have.

\n

\r

\"

\'

\\

...

header files

Ex) import sympy as sp

⇒ libraries

libraries

stdio.h (≠ studio.h.)

Manual Pages

manual.cs50.io

manual.cs50.io/#stdio.h

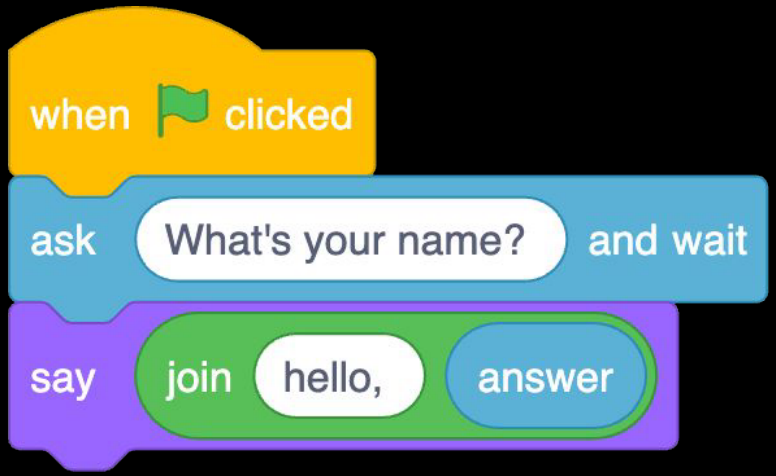
manual.cs50.io/3/printf

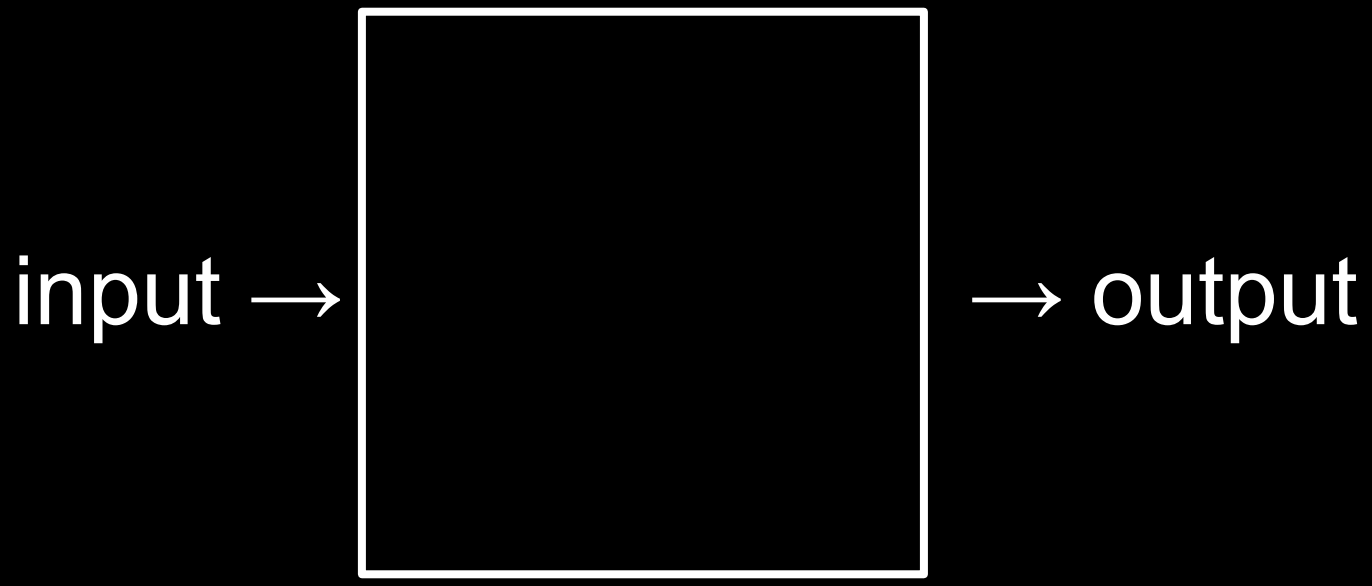
cs50.h

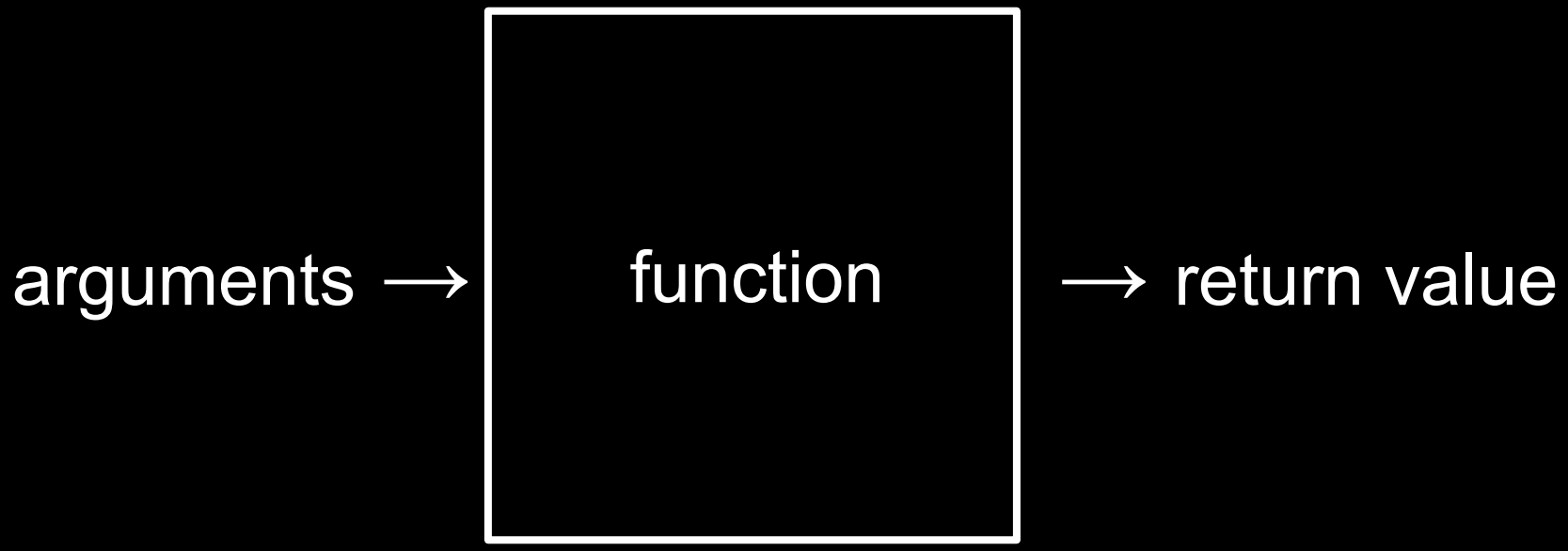
manual.cs50.io/#cs50.h

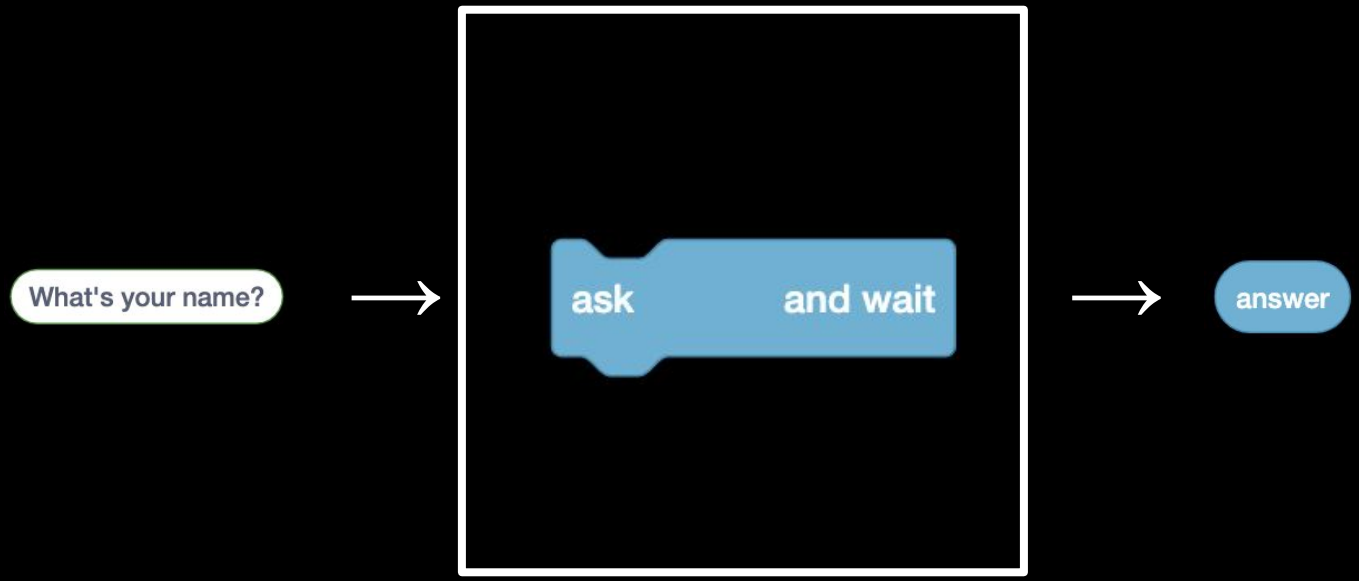
```
get_char

get_double

get_float

get_int

get_long

get_string

...
```

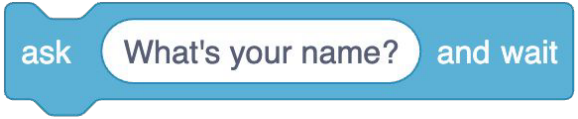
```
get_char

get_double

get_float

get_int

get_long

get_string

...
```


when clicked
ask What's your name? and wait
say join hello, answer

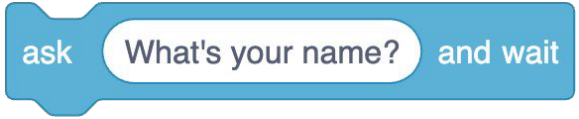

input →
→ output


arguments
function
return value

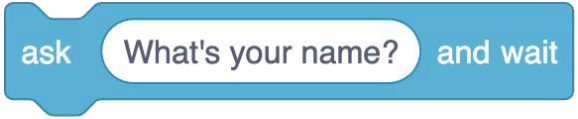

What's your name?
ask
and wait
answer


ask
What's your name?
and wait

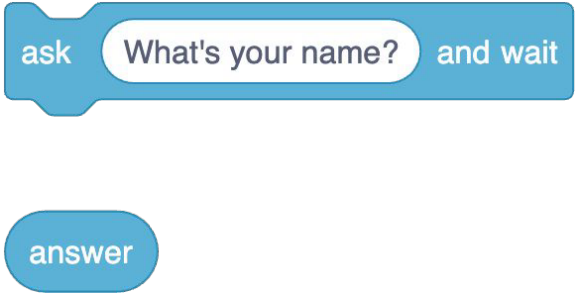
answer



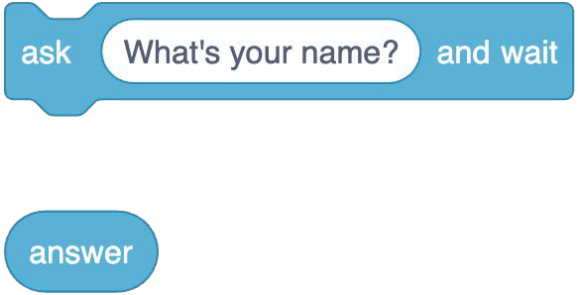
```
get_string(                    )
```



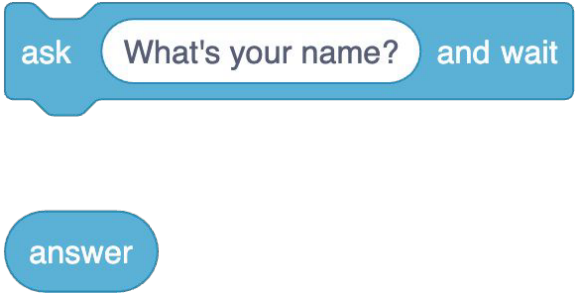
```
get_string( What's your name?  )
```



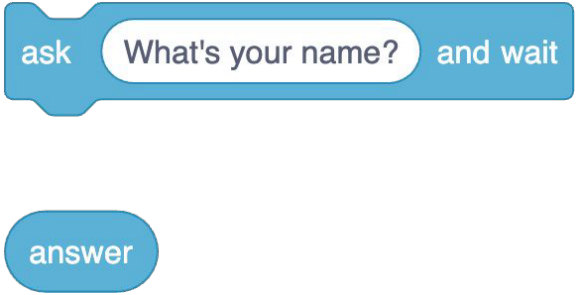

```
get_string("What's your name? ")
```




```
answer = get_string("What's your name? ")
```




```
string answer = get_string("What's your name? ")
```




```
string answer = get_string("What's your name? ");
```

say
join
hello,
answer

```
printf(                    );
```

```
printf( hello, %s          );
```

```
printf("hello, %s  "        );
```

```
printf("hello, %s\n"        );
```

```
printf("hello, %s\n", answer);
```

Linux

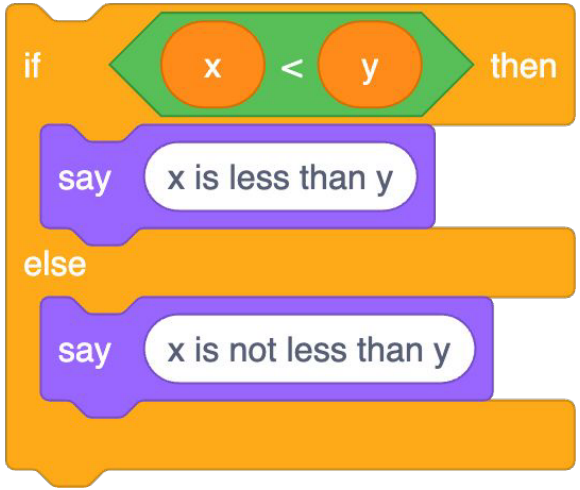
```
cd
cp
ls
mkdir
mv
rm
rmdir
...
```

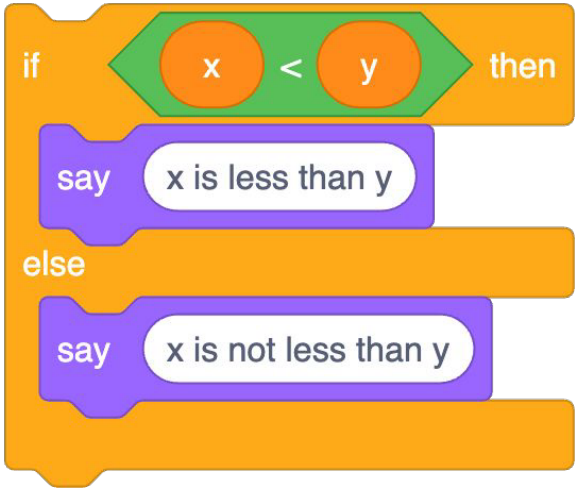
conditionals

if
x
<
y
then
say
x is less than y

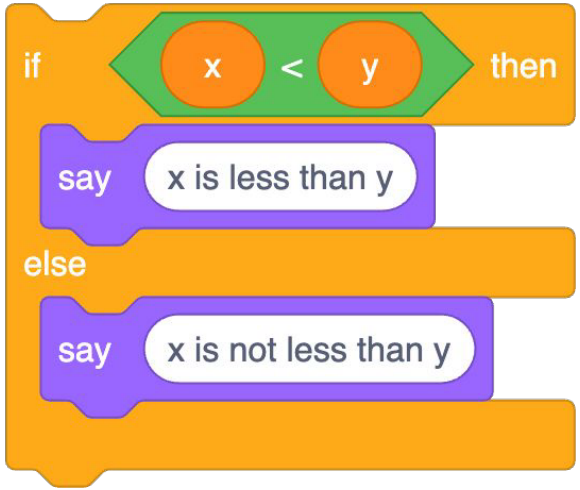
```
if (x < y)
{

}
```

```
if (x < y)
{
    printf("x is less than y\n");
}
```


if
x
<
y
then
say
x is less than y
else
say
x is not less than y

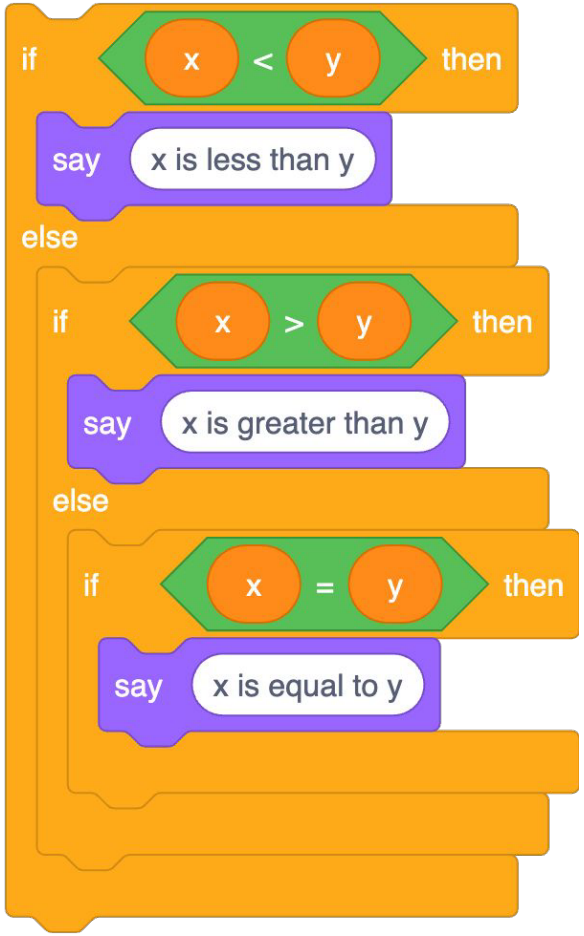



```
if (x < y)
{

}
else
{

}
```

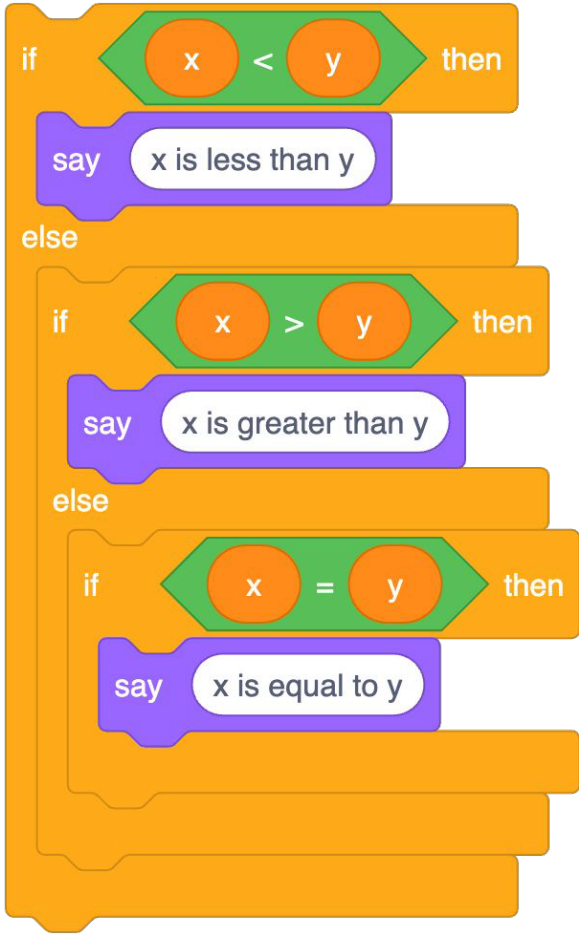



```
if (x < y)
{
    printf("x is less than y\n");
}
else
{
    printf("x is not less than y\n");
}
```

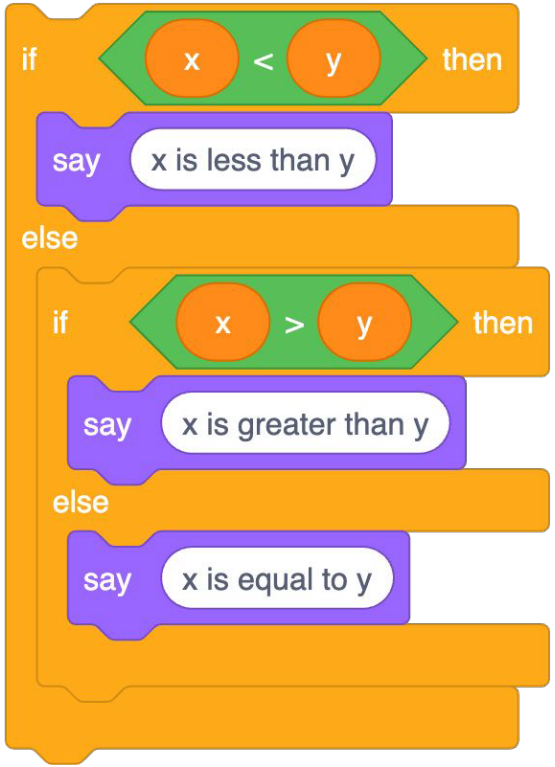
```
if <x < y> then
  say [x is less than y]
else
  if <x > y> then
    say [x is greater than y]
  else
    if <x = y> then
      say [x is equal to y]
```




```
if (x < y)
{

}
else if (x > y)
{

}
else if (x == y)
{

}
```




```
if (x < y)
{
    printf("x is less than y\n");
}
else if (x > y)
{
    printf("x is greater than y\n");
}
else if (x == y)
{
    printf("x is equal to y\n");
}
```




```
if (x < y)
{
    printf("x is less than y\n");
}
else if (x > y)
{
    printf("x is greater than y\n");
}
else
{
    printf("x is equal to y\n");
}
```

operators

=

<

<=

>

>=

==

!=

...

types

```
bool

char

double

float

int

long

string

...
```

bool

char

double

float

int

long

string

...

cs50.h

```
get_char

get_double

get_float

get_int

get_long

get_string

...
```

```
get_char

get_double

get_float

get_int

get_long

get_string

...
```

format codes

%c

%f

%i

%li

%s

%c

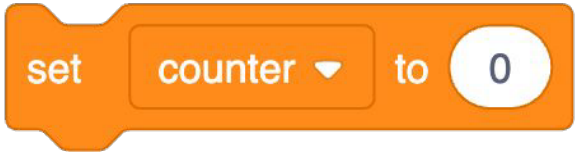
%f

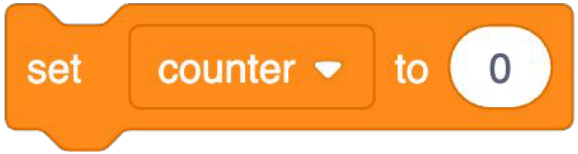
%i

%li

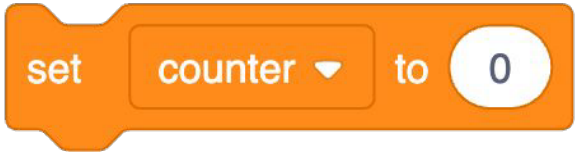
%s

variables

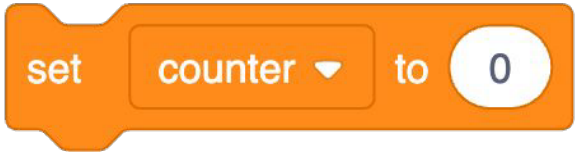

set
counter
to
0



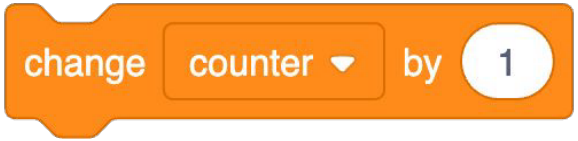

```
counter = 0
```

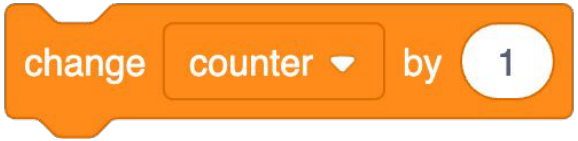



```
int counter = 0
```

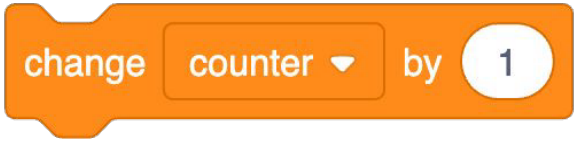



```
int counter = 0;
```

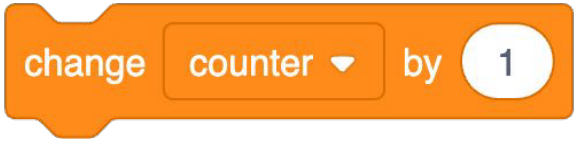

change
counter
by
1



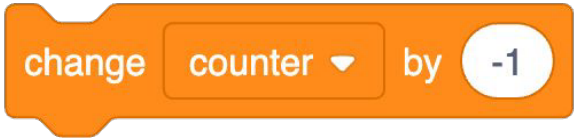

```
counter = counter + 1;
```



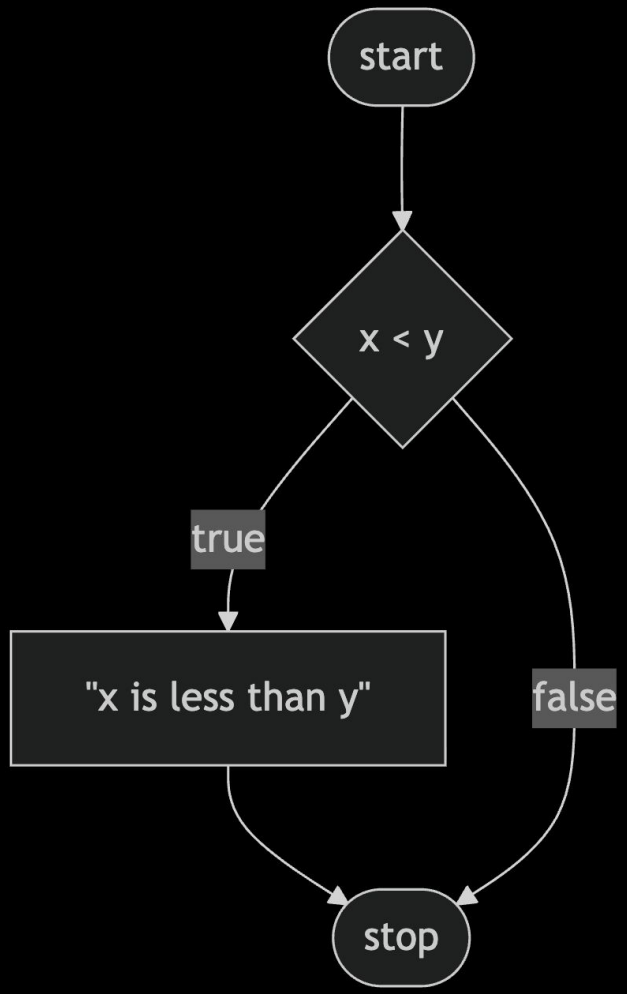

```
counter += 1;
```

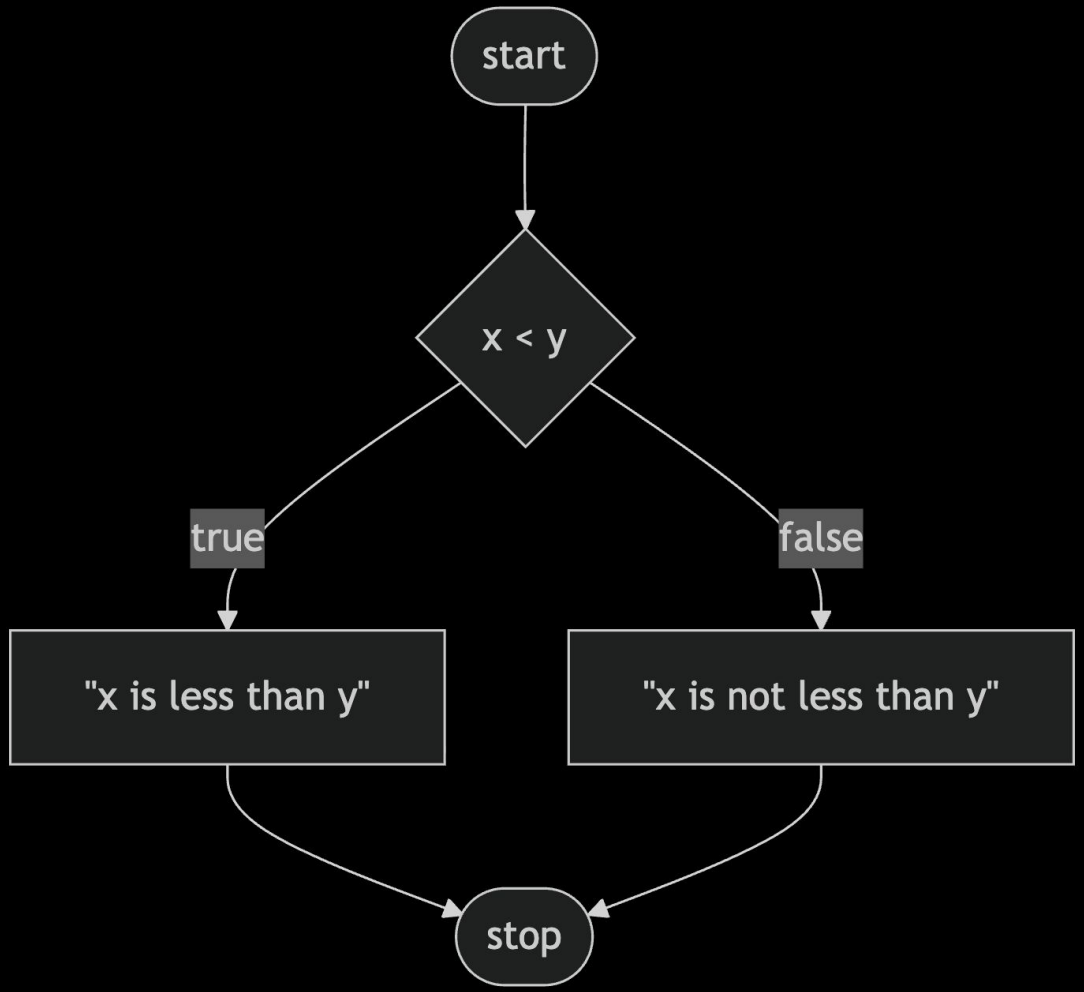


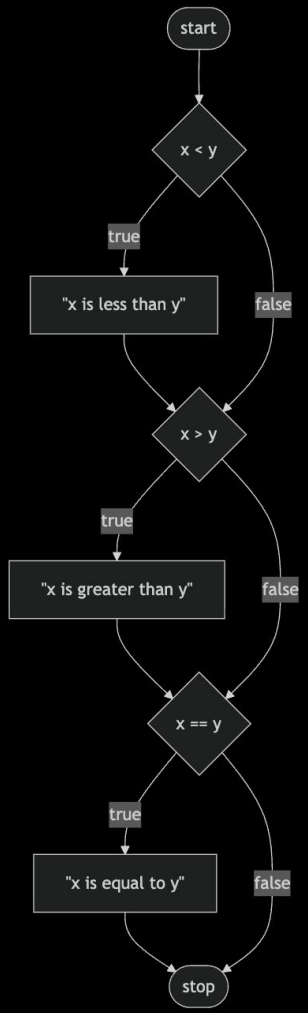

```
counter++;
```

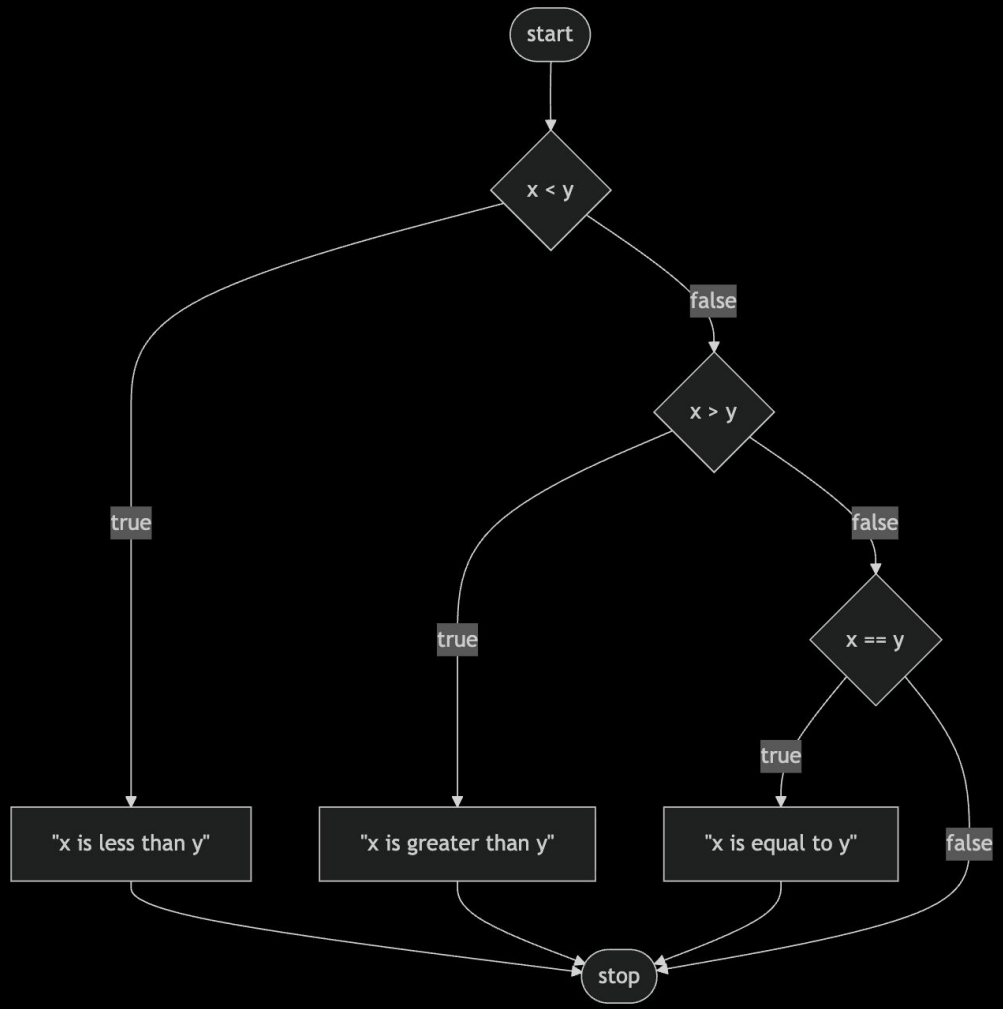


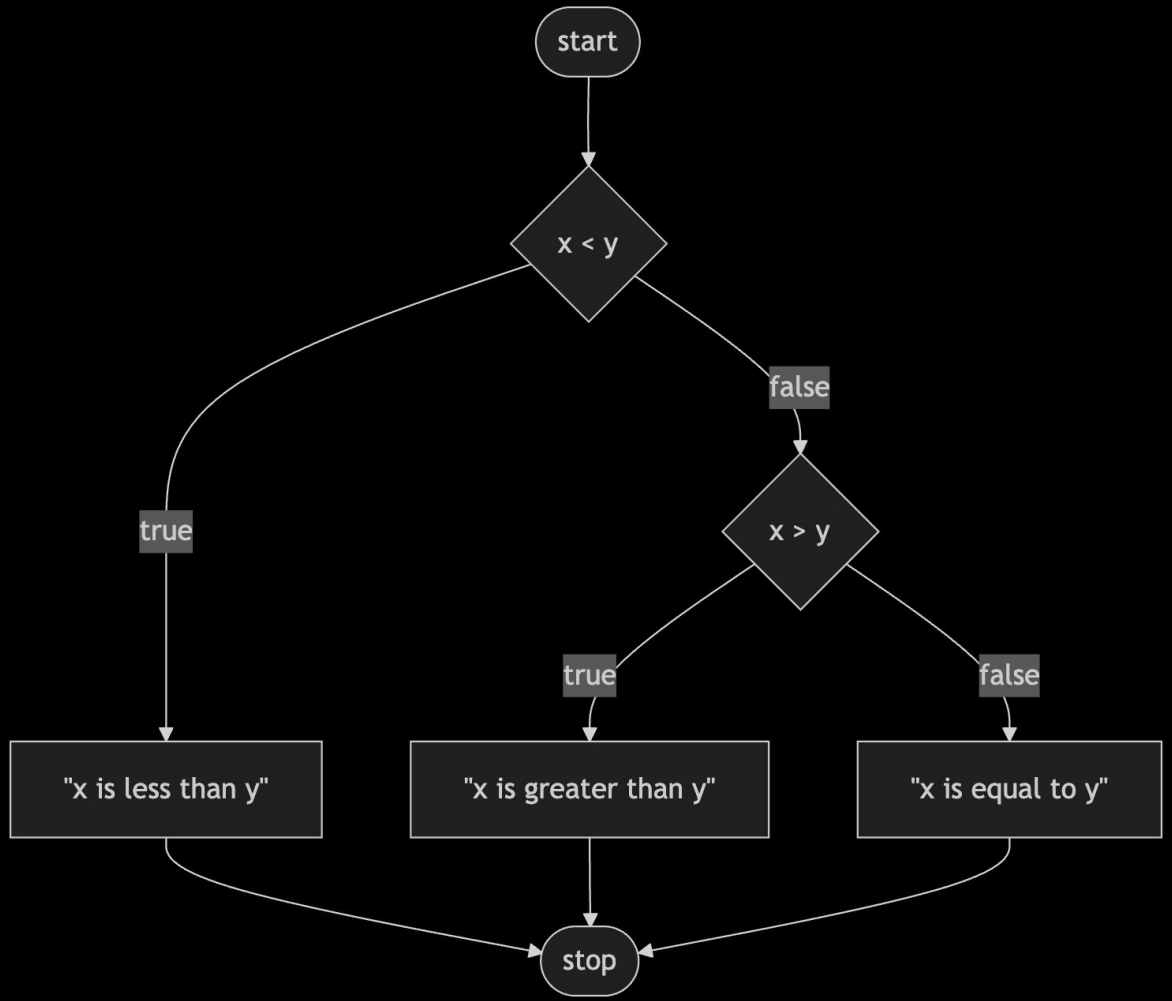

```
counter--;
```


start
x < y
true
"x is less than y"
false
stop


start
x < y
true
false
"x is less than y"
"x is not less than y"
stop


start
x < y
true
"x is less than y"
false
x > y
true
"x is greater than y"
false
x == y
true
"x is equal to y"
false
stop


start
x < y
true
false
x > y
true
false
x == y
true
false
"x is less than y"
"x is greater than y"
"x is equal to y"
stop


start
x < y
true
false
x > y
true
false
"x is less than y"
"x is greater than y"
"x is equal to y"
stop

bool

char

double

float

int

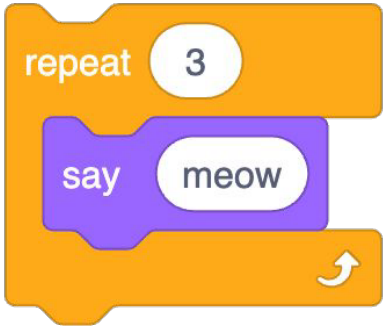
long

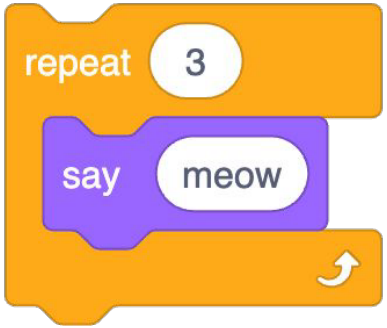
string

...

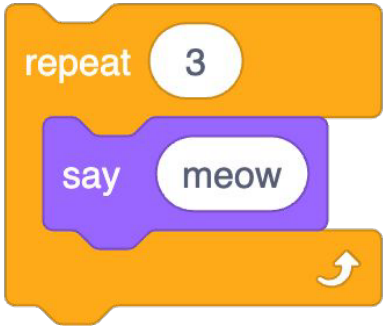
```
get_char

get_double

get_float

get_int

get_long

get_string

...
```

loops

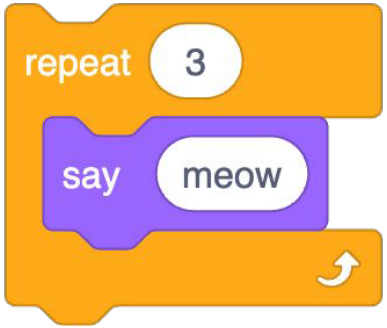

repeat
3
say
meow



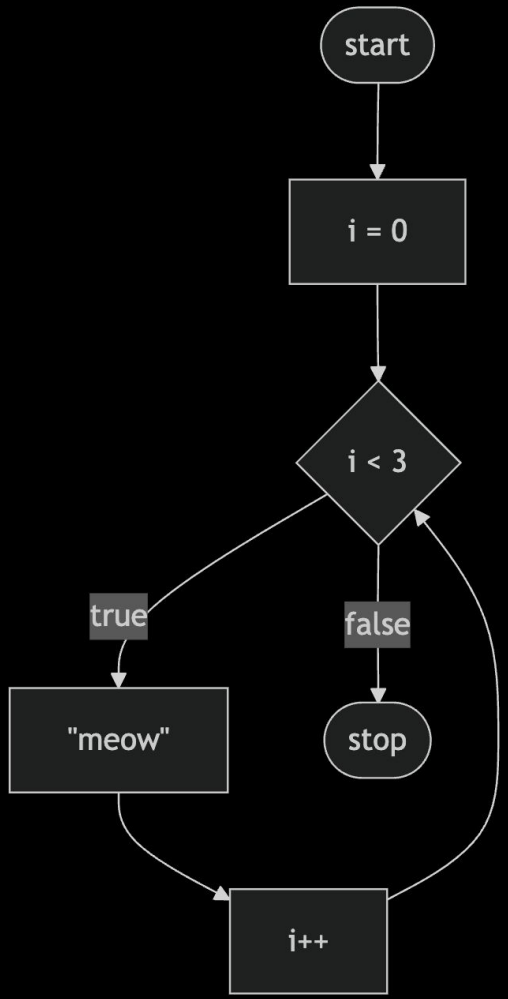

```
int i = 3;
while (i > 0)
{

    i--;
}
```

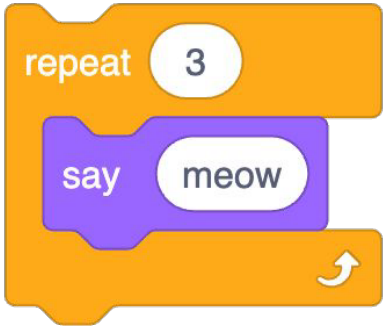



```
int i = 3;
while (i > 0)
{
    printf("meow\n");
    i--;
}
```

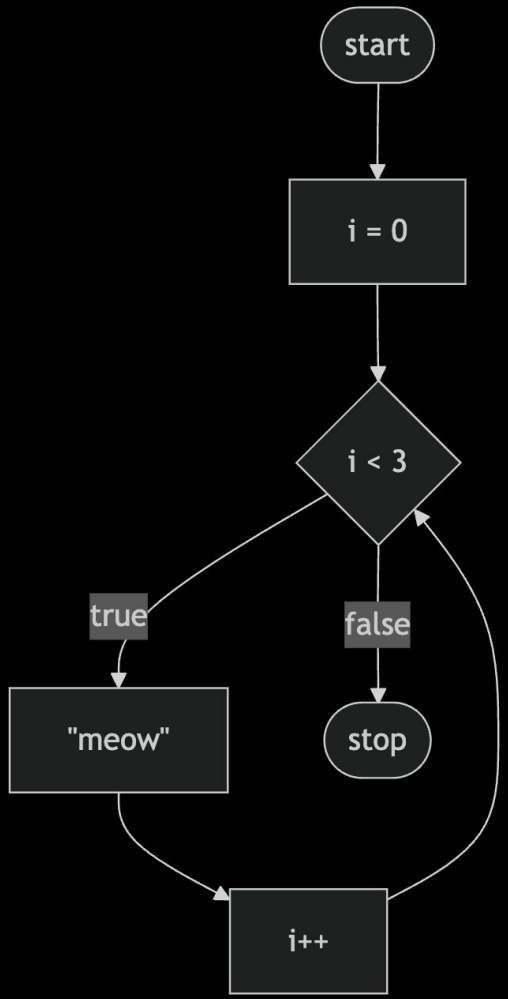


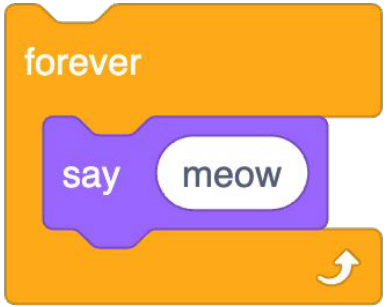

```
int i = 0;
while (i < 3)
{
    printf("meow\n");
    i++;
}
```

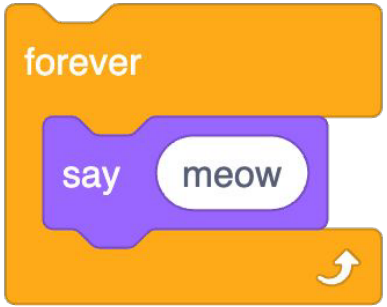

start
i = 0
i < 3
true
false
"meow"
stop
i++



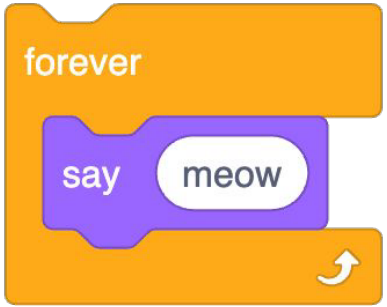

```
for (int i = 0; i < 3; i++)
{
    printf("meow\n");
}
```


start
i = 0
i < 3
true
false
"meow"
stop
i++

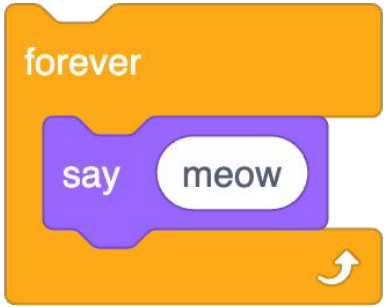

forever
say
meow



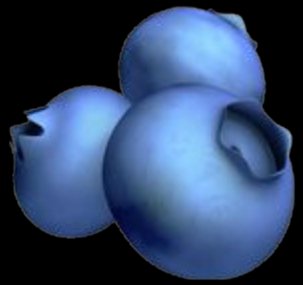

```
while (    )
{

}
```




```
while (true)
{

}
```

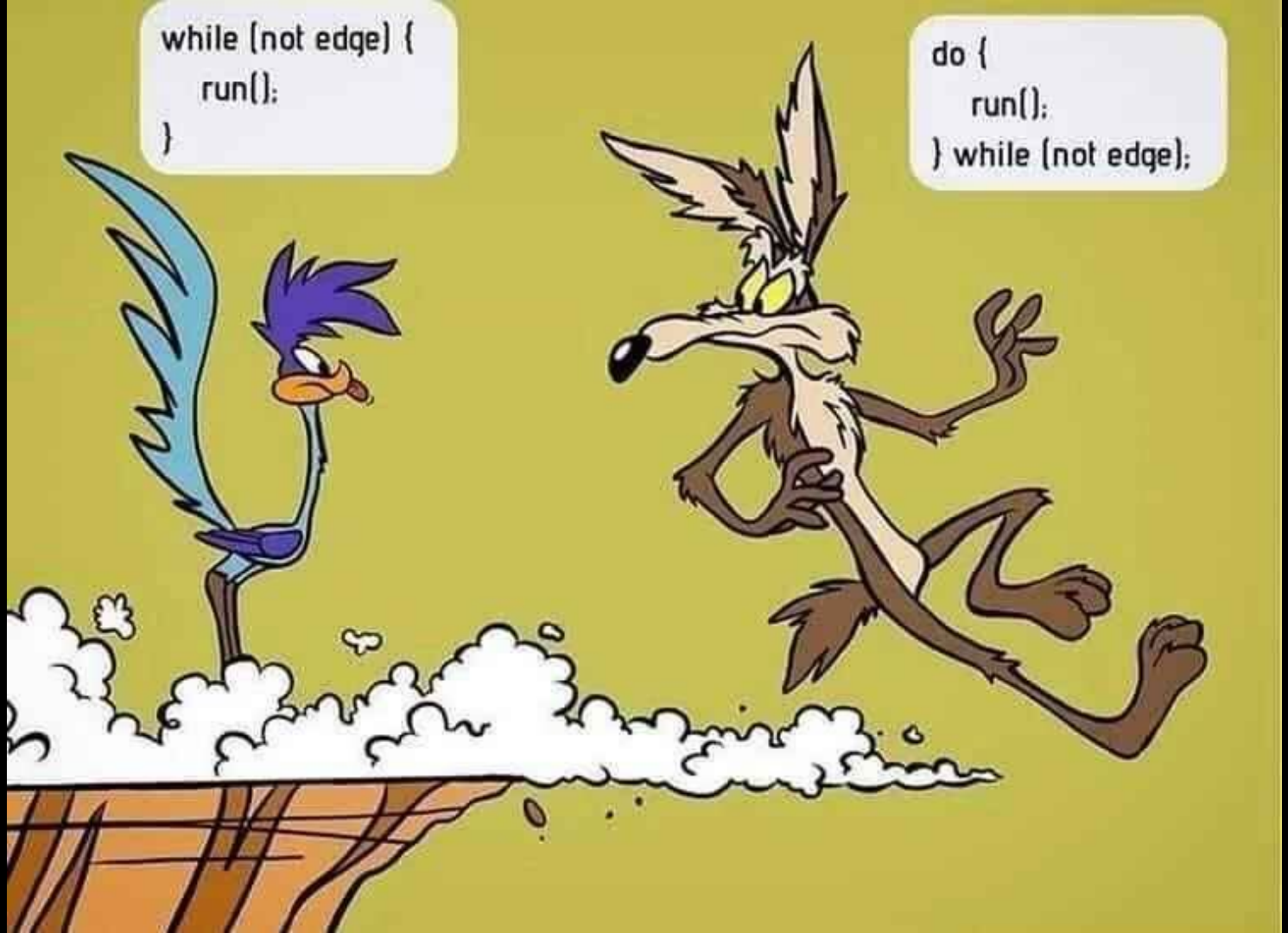


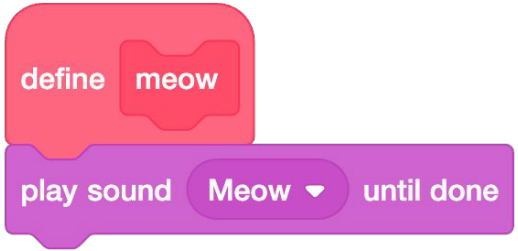

```
while (true)
{
    printf("meow\n");
}
```

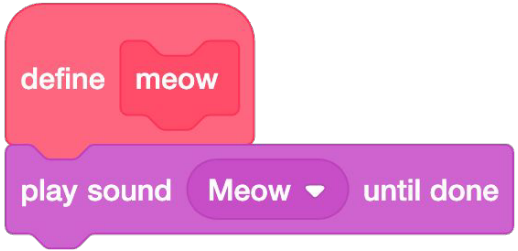


control-c

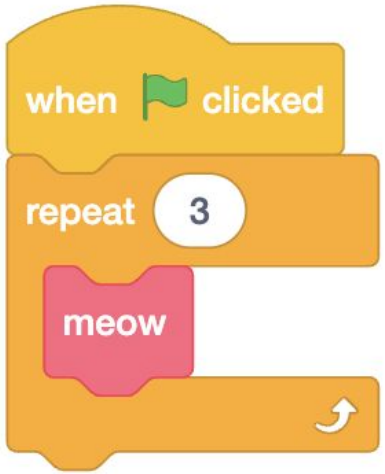
scope

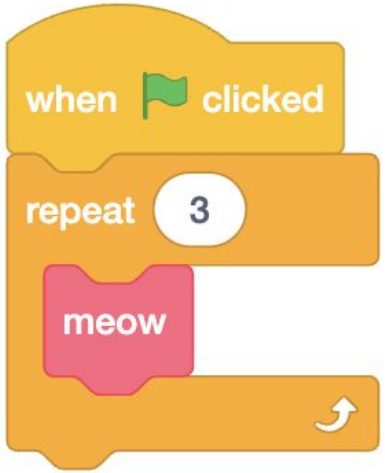

while (not edge) {
run();
}
do {
run();
} while (not edge);


define meow
play sound Meow until done

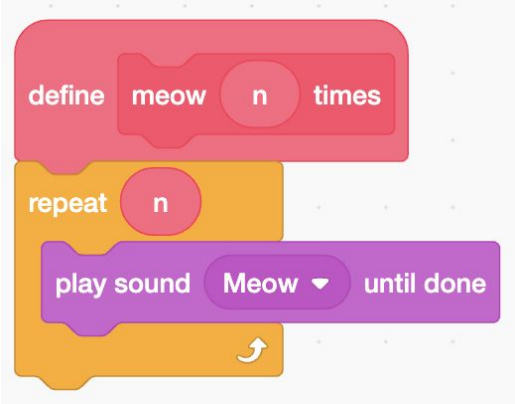


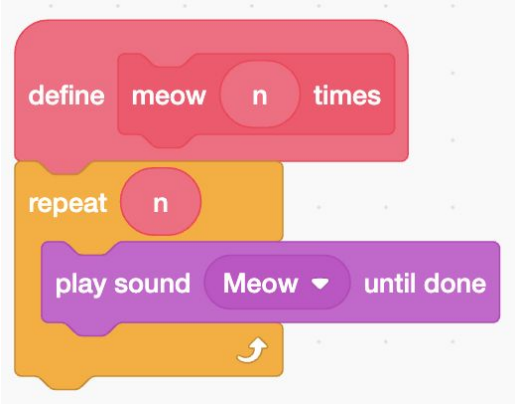

```
void meow(void)
{
    printf("meow\n");
}
```


when clicked
repeat 3
meow

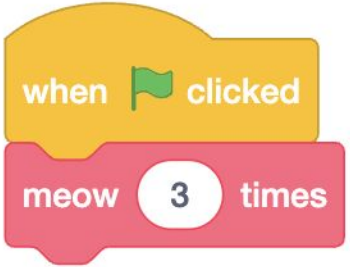


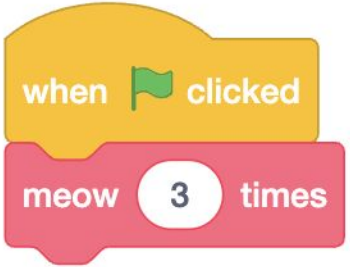

```
int main(void)
{
    for (int i = 0; i < 3; i++)
    {
        meow();
    }
}
```


define meow n times
repeat n
play sound Meow until done




```
void meow(int n)
{
    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```


when clicked
meow 3 times



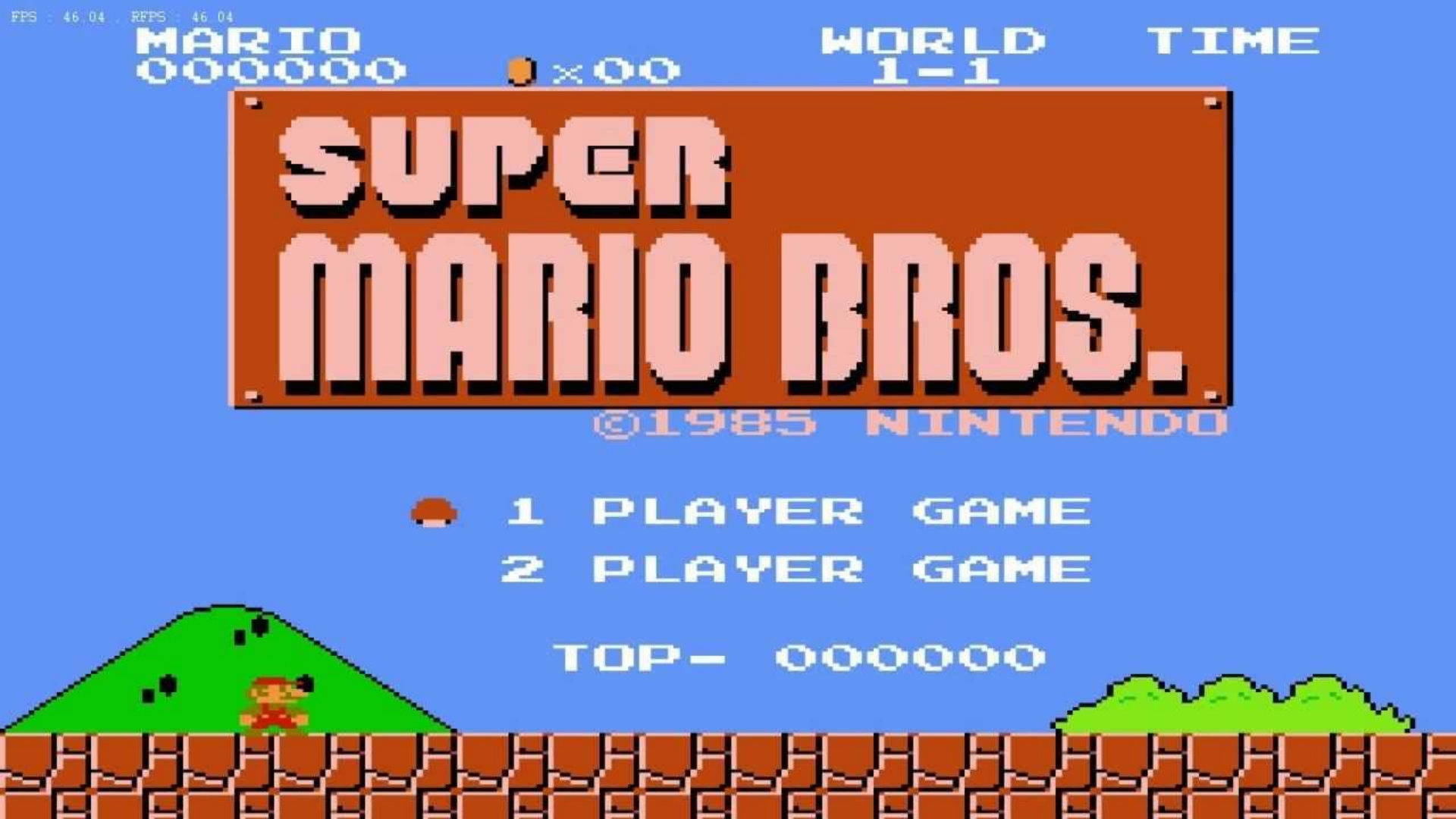

```
int main(void)
{
    meow(3);
}
```

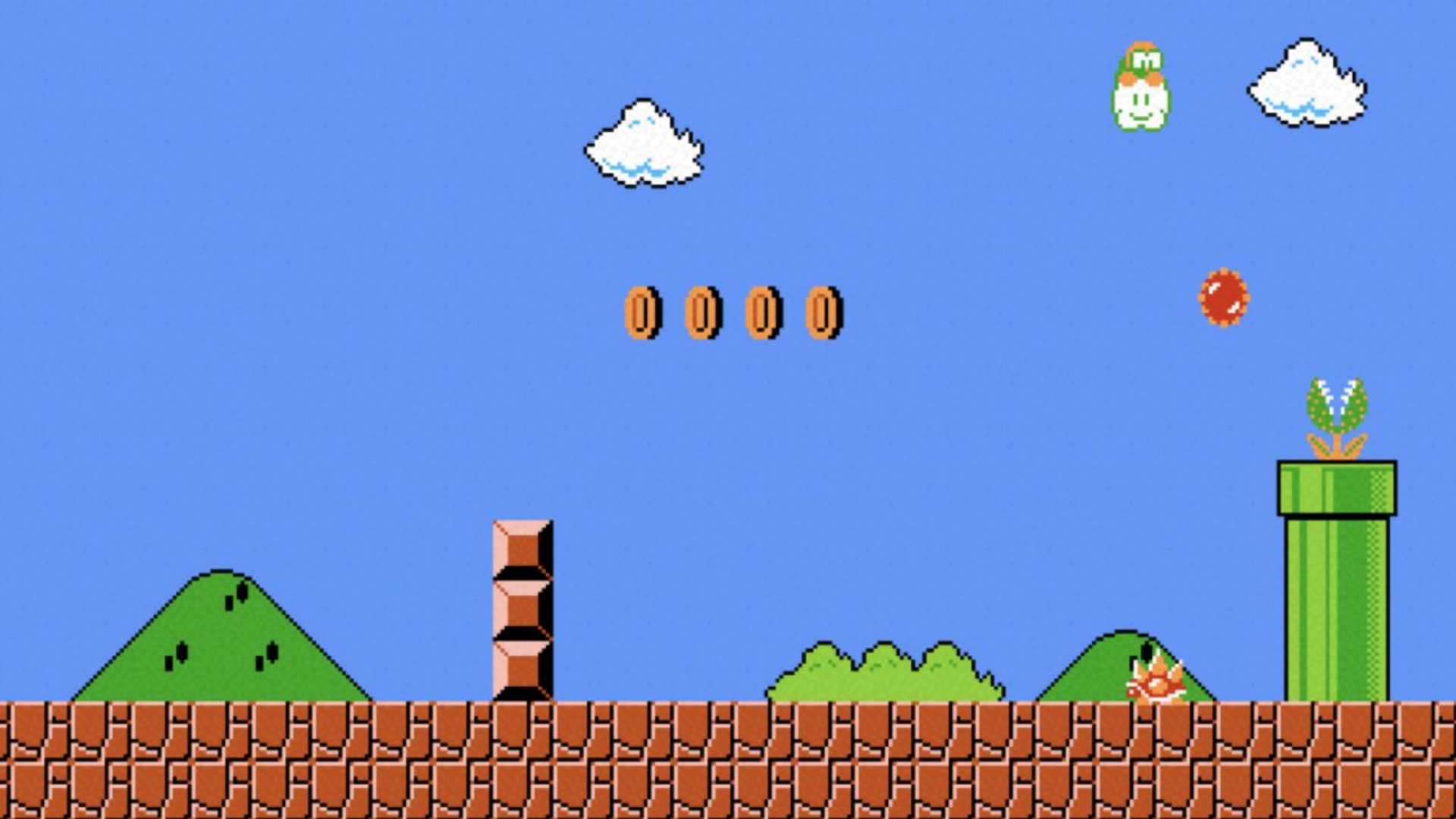
scope

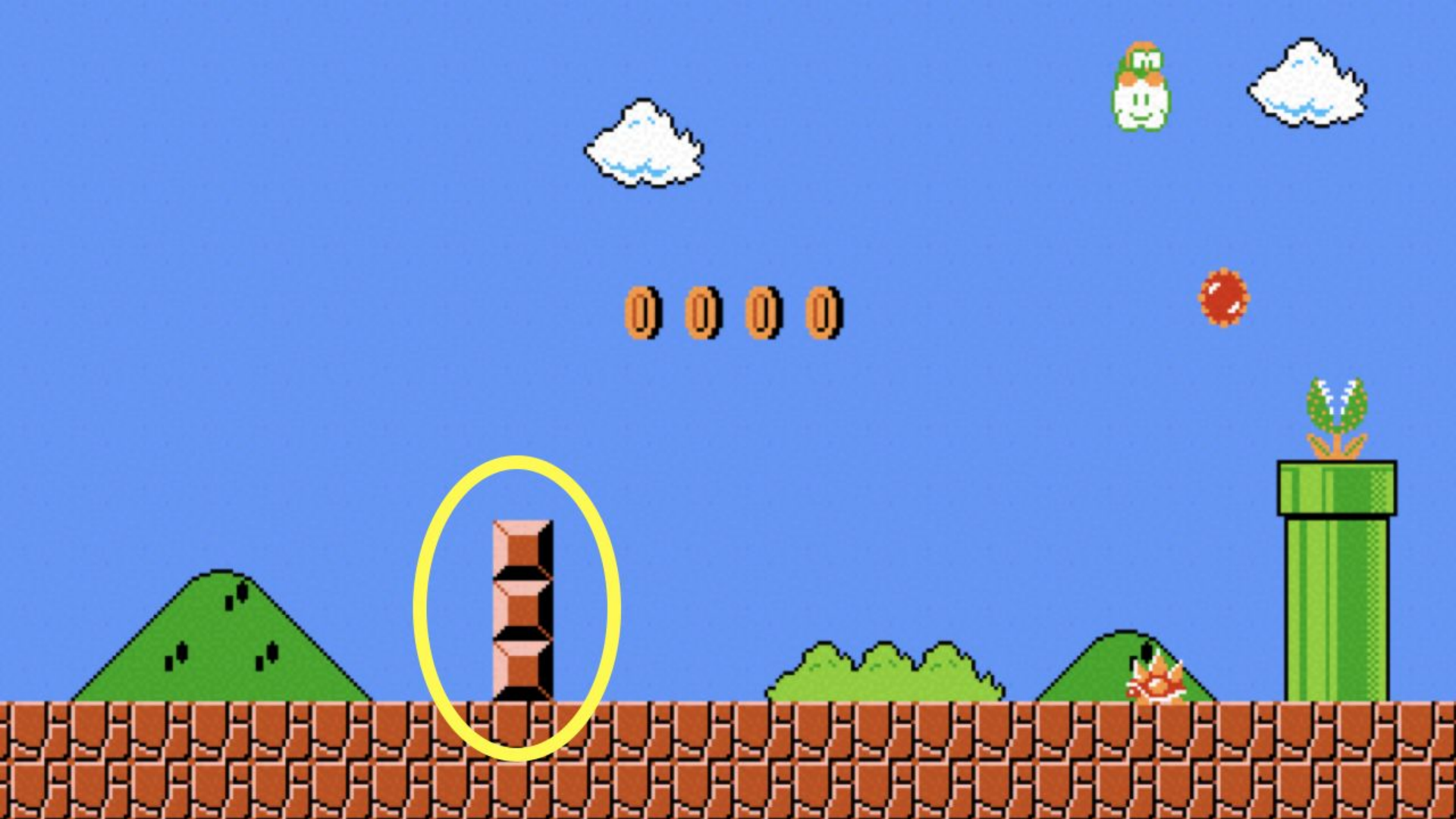
correctness, design, style

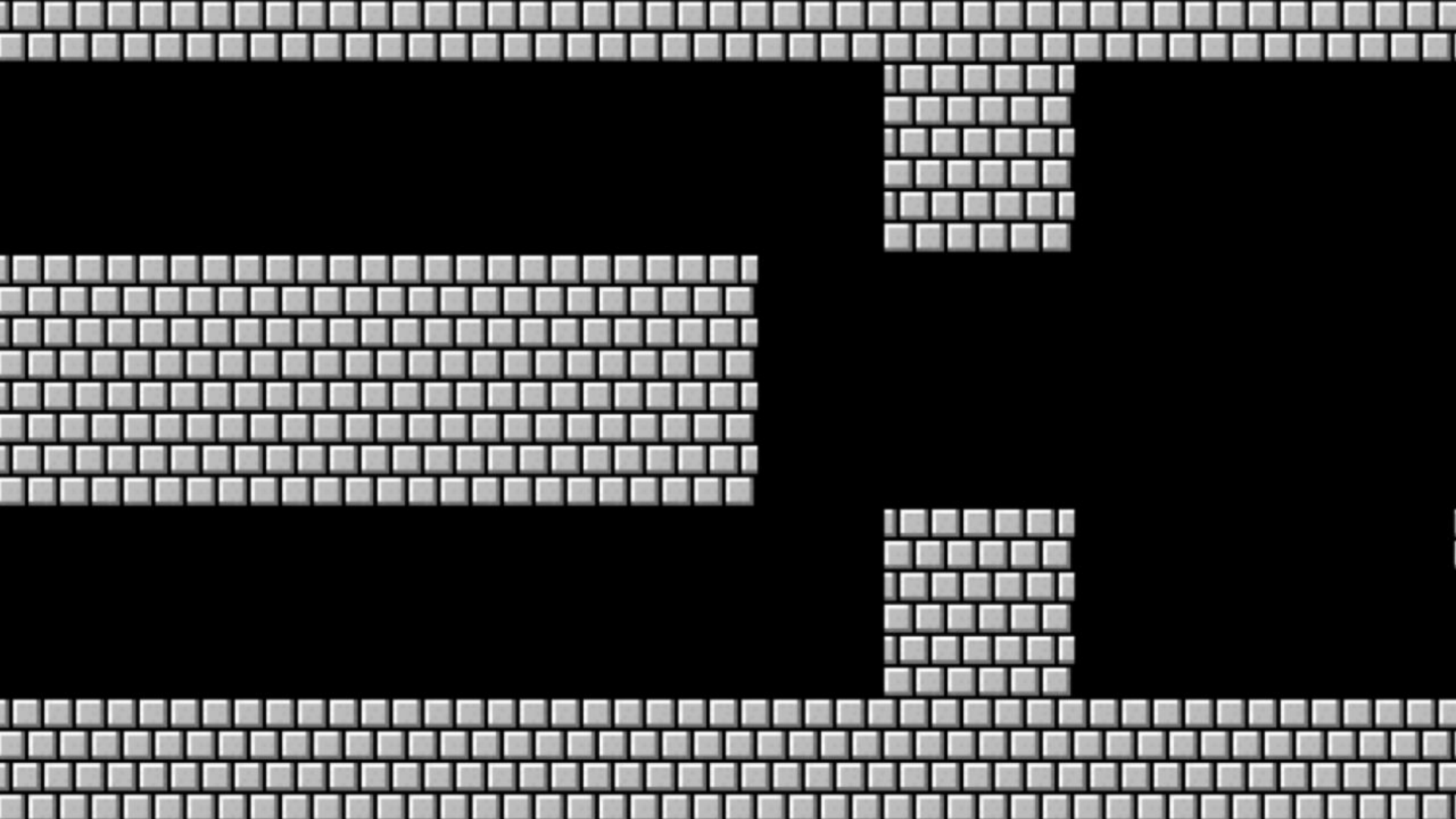
check50, design50, style50

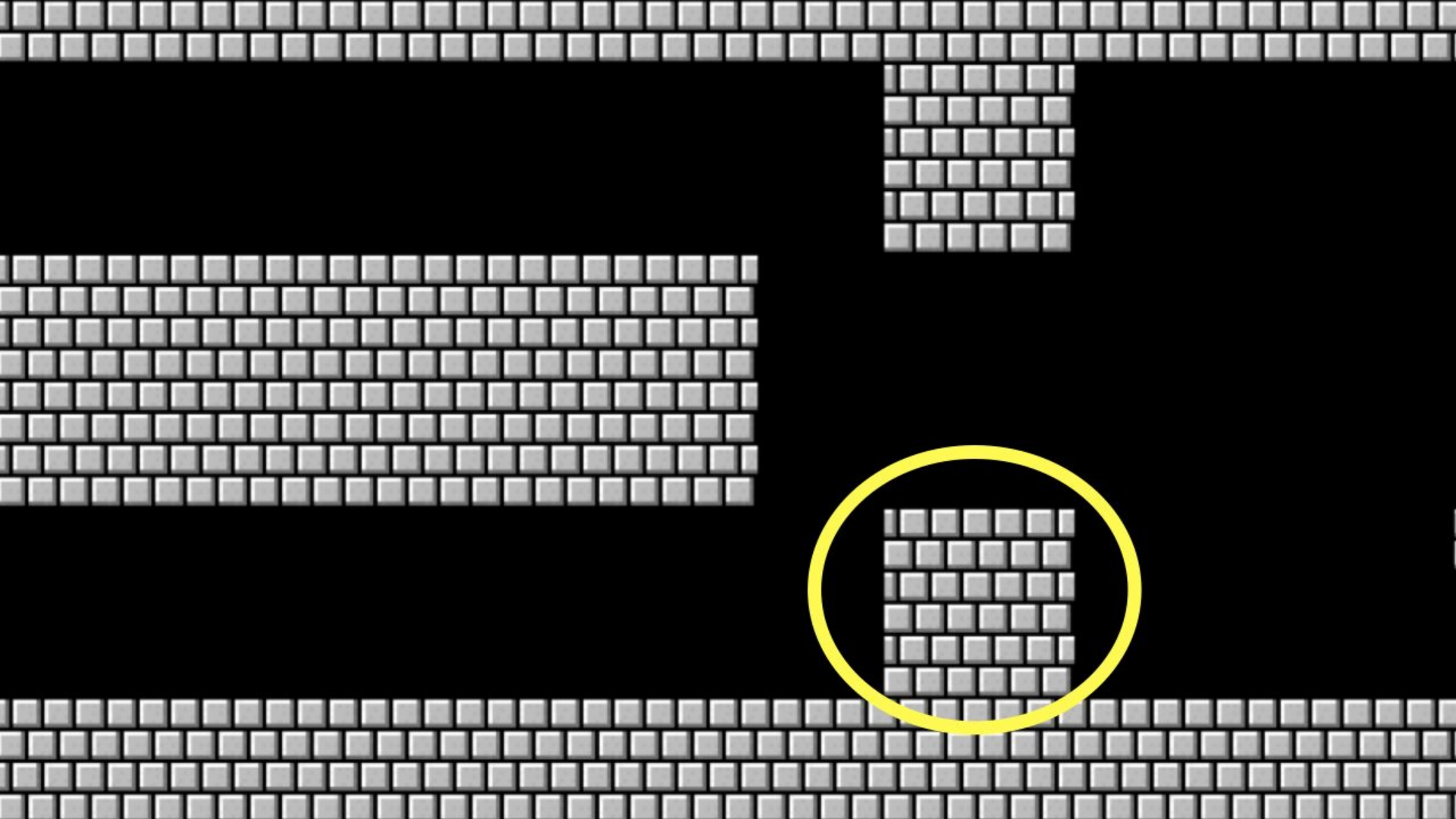
arguments $\longrightarrow$ function $\longrightarrow$ side effects


FPS : 46.04 . RFPS : 46.04
MARIO
000000
×00
WORLD
1-1
TIME
SUPER
MARIO BROS.
©1985 NINTENDO
1 PLAYER GAME
2 PLAYER GAME
TOP- 000000

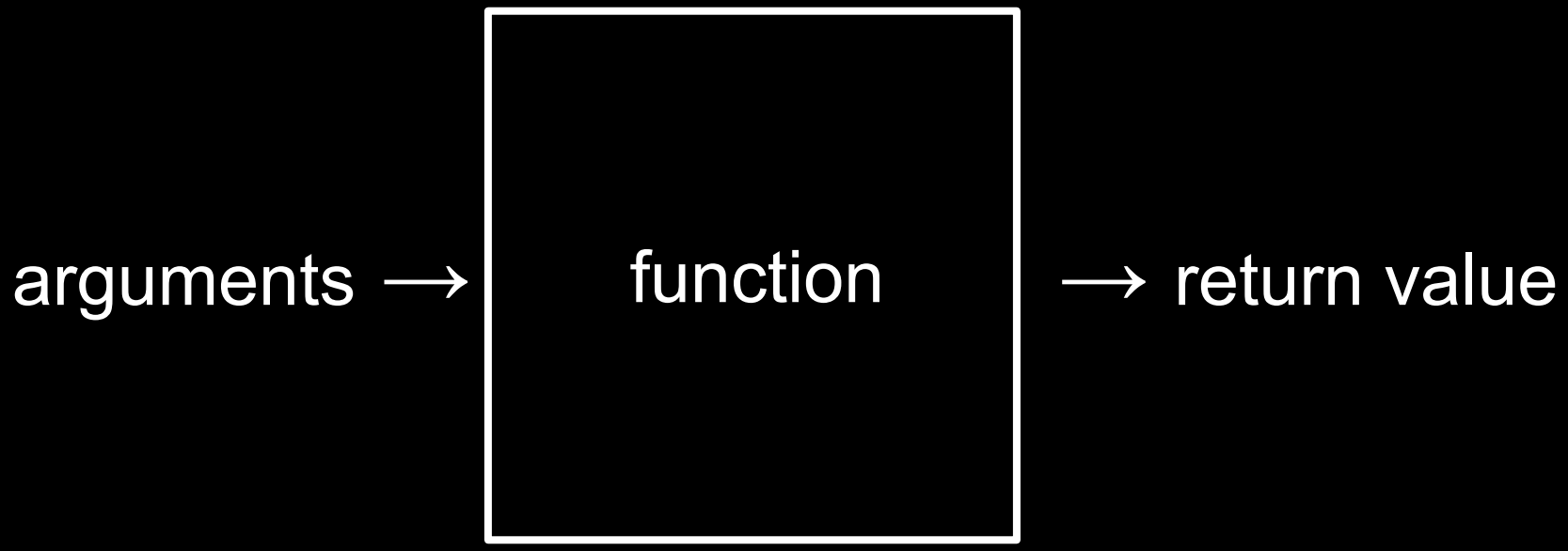








constants


arguments
function
return value

+

-

*

/

%

...

integer overflow

crucial
by Micron
47J6
4G85
8BB12
D9HXT
4G85
8BB12
D9HXT
4G85

amazon.com

000

001

010

011

100

101

110

111

1000

000

Using MAME to warp to level 256, the split screen is shown.

truncation

floating-point imprecision

Year 2000 Problem

1999

1999

1900

Year 2038 Problem

19 January 2038

13 December 1901

19 January 2038

This is CS50

This is CS50

This is CS50

This is CS50

This is CS50

This is CS50

This is CS50

This is CS50

This is CS50